



2ND HACKATHON CAMEROONIAN GIRLS IN STEAM · GIRLS IN CYBER DEFENSE

PhishLense

A prototype for detecting smishing now, expanding next to phishing links, websites, and deepfakes.

 *Team: Genesys · Challenge: Girls in Cyber Defense · Members: Serge, Noel, Scripture, Aliya*

WITH THE SUPPORT OF OUR PARTNERS



The problem we are solving

⚠ *Phishing has moved beyond email — into SMS, links, fake websites, and deepfakes.
Users react fast, because scam messages look urgent, familiar, or trusted.*

In Cameroon, online scams and phishing continue to cause major financial losses.

💡 *A striking figure:*

**1.027 billion
CFA**

lost to online scams in Cameroon in 2025.

Our solution



In one sentence:

PhishLense is a working cybersecurity prototype that detects smishing messages and is designed to grow into phishing link and deepfake detection.



Benefit 1:

Warns users before they click, reply, or share sensitive information.



Benefit 2:

Gives a label, a confidence score, and a simple explanation.



Benefit 3:

Built for real users, not only for technical demonstrations.

03 · DEMONSTRATION

Here is our prototype

The screenshot displays a web-based prototype for SMS analysis. On the left, a sidebar contains 'Demo controls' with a 'Sample SMS' dropdown set to 'Safe family message' and a 'Safe vs scam' toggle. Below this is a 'Quick samples' section with five message snippets. The main area shows an input field with a sample message about a blocked Orange Money account, an 'Analyze SMS' button, and a 'Clear Input' button. Below the input, a 'Smishing / Scam' alert is shown with a 'Confidence - 100%' bar and an 'Estimated user-facing risk level' bar. To the right, a 'Class probabilities' bar chart shows 'smishing' as the most likely class. Further right, a 'Why this was flagged' section lists five reasons: 'Contains a link', 'Requests secret credentials', 'Uses urgency or account-pressure language', and 'Uses threat or fear language', each with a 'HIGH' risk level. A third section on the right lists three steps: '1. Select or Paste Input', '2. Run Vector Classification', and '3. Evaluate Risk Metrics'.

Demo controls

Sample SMS

Safe family message

Safe vs scam

Quick samples

- Safe family message Hi mum, I will be home by 6pm. Please keep di...
- Promotional spam FREE airtime for all users this weekend. Text...
- Orange Money scam Orange Money account blocked. Verify immediat...
- French banking scam Votre compte bancaire sera suspendu. Confirme...
- MTN MoMo PIN scam MTN MoMo: Suspicious login detected. Send you...

Orange Money account blocked. Verify immediately: <https://om-secure-login.com>

Analyze SMS Clear Input

Smishing / Scam

Recommended action: Do not click links, do not share OTP/PIN/passwords, and verify through the official app, branch, or known customer-care number.

Confidence - 100%

Estimated user-facing risk level

Class probabilities

Class	Probability
ham	0.0
smishing	1.0
spam	0.0

Why this was flagged

- Contains a link** (HIGH)
Scam SMS often push users to fake login or payment pages.
- Requests secret credentials** (HIGH)
Legitimate services should not ask for PIN, OTP, password, or card details by SMS.
- Uses urgency or account-pressure language** (HIGH)
Urgent deadlines reduce careful checking and are common in smishing.
- Uses threat or fear language** (HIGH)
Warnings about blocked accounts, accidents, or failed payments can manipulate users.

1. Select or Paste Input (STEP 01)
Choose a local sample from the sidebar or paste custom SMS text.

2. Run Vector Classification (STEP 02)
Click Analyze SMS to calculate prediction weights and probabilities.

3. Evaluate Risk Metrics (STEP 03)
Review the explicit risk rating, linguistic flags, and safety advice.



The highlight moment

- Paste a message.
- Show the prediction.
- Show the explanation.
- Use the backup screenshot if needed.



Our cybersecurity component

🔒 *AI-based detection flags suspicious SMS, fake URLs, and phishing sites in real time.*

Warns users before they click, log in, or send money — stopping social engineering in its tracks.

Explains clearly why a message, link, or site is risky.

Teaches users over time — every alert builds scam awareness.

For whom, and what impact?

For whom?

PhishLense is for mobile users, students, families, and small businesses.

Also built for developers and organizations integrating our API.

This expands the product from a simple app into a reusable security tool.

What impact?

PhishLense warns users before they click, reply, or pay, helping stop scams early.

Clear risk signals teach users to spot scams themselves.

This can reduce fraud, protect money, and improve digital trust in everyday phone use.

Viable and unique

💰 *How is it adopted / funded?*
Free app for individuals.

Paid API access for developers and organizations.

Revenue from subscriptions, enterprise licenses, and custom integrations.

This makes PhishLense both a standalone app and a security layer inside other systems.

★ *What makes you unique?*

Trained on Cameroon-relevant data.

Built to support bilingual use.

Detects SMS scams, fake URLs, phishing websites, and future deepfake threats.

Helps users learn by explaining why content is suspicious.

07 · THE TEAM

Who we are



*Serge
Coordinator*



*Noel
Developer*



*Scripture
Model Training*



*Aliya
Data Mining*



08 · OUR ASK

Thank you!

🤝 *We're seeking mentoring, partnership, incubation, and support to grow this prototype into a real product.*

Contact: phishlense@gmail.com · +237671160734

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